

Problem

Sketch the Bode plots for

$$\mathbf{H}(\omega) = \frac{50 j\omega}{(j\omega + 4)(j\omega + 10)^2}$$

Step-by-step solution

Step 1 of 7

Given transfer function is

$$H(\omega) = \frac{50 j\omega}{(j\omega + 4)(j\omega + 10)^2}$$

$$H(\omega) = \frac{50 j\omega}{400 \left(1 + \frac{j\omega}{4}\right) \left(1 + \frac{j\omega}{10}\right)^2}$$

$$H(\omega) = \frac{j\omega}{8 \left(1 + \frac{j\omega}{4}\right) \left(1 + \frac{j\omega}{10}\right)^2}$$

$$\Rightarrow H(\omega) = \frac{|j\omega|}{8 \left|1 + \frac{j\omega}{4}\right| \left|1 + \frac{j\omega}{10}\right|^2} \angle 90^\circ - \tan^{-1}\left(\frac{\omega}{4}\right) - 2 \tan^{-1}\left(\frac{\omega}{10}\right)$$

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Hence, the magnitude and phase are,

$$H_{dB} = 20 \log_{10}(\omega) - 20 \log_{10} 8 - 20 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{4}\right)^2} \right) - 40 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{10}\right)^2} \right) \dots\dots (1)$$

$$\phi = 90^\circ - \tan^{-1}\left(\frac{\omega}{4}\right) - 2 \tan^{-1}\left(\frac{\omega}{10}\right) \dots\dots (2)$$

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The various terms of $H(\omega)$ are listed in the increasing order of their Corner frequencies. Also to table shows to slope contributed by each term and the change in slope at the Corner frequency.

Term	Corner Frequency (rad/sec)	Slope dB/sec	Change in slope dB/dec
$j\omega$	—		
$\frac{1}{1 + \frac{j\omega}{4}}$	$\omega = 4$	+20	$+20 - 20 = 0$
$\frac{1}{1 + \frac{j\omega}{10}}$	$\omega = 10$	-20	$0 - 20 = -20$

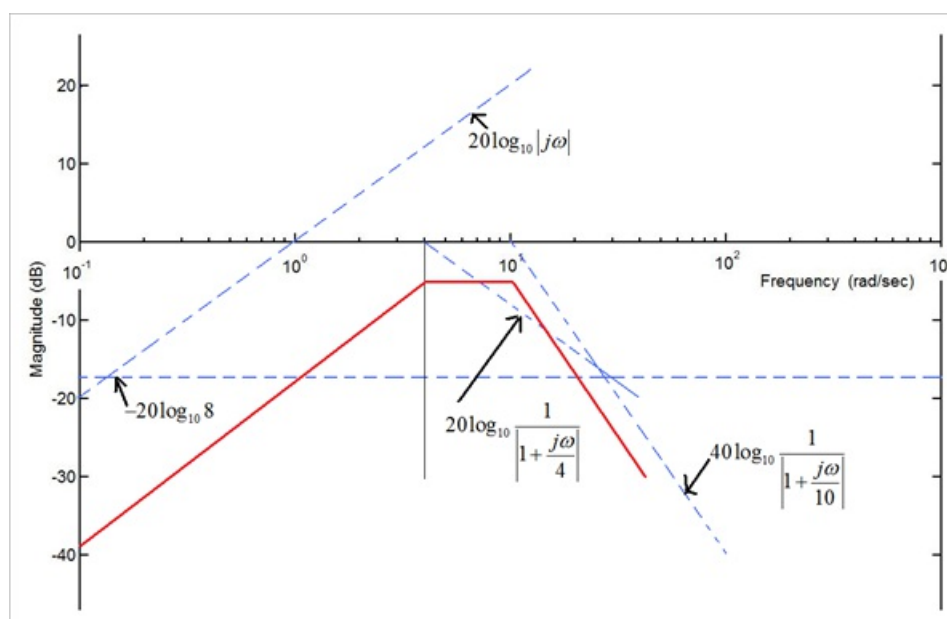
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Below table shows gain values for different values of ' ω ' according to equation (1)

ω	H_{dB}
0.1	-38
4	-10
10	-10

The magnitude plot is

Step 5 of 7



Step 6 of 7

The phase angle of ϕ are calculated for various values of ' ω ' and listed in below

$$\phi = 90^\circ - \tan^{-1}\left(\frac{\omega}{4}\right) - 2 \tan^{-1}\left(\frac{\omega}{10}\right)$$

ω	$\phi(\text{deg})$
1	64.5
40	-146

The phase plot is

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